WEAVE: Wavelet-Routed Latent Style Transfer Trained in Minutes

Anonymous submission

Abstract

Current style transfer methods are often ranked below a trivial identity mapping under commonly reported metrics such as LPIPS and ArtFID, which primarily reward content preservation. On direct style-similarity metrics, some spatial-domain models trained for hours even fall below this identity baseline. While state-of-the-art diffusion models and training-free methods built on them achieve stronger target-style transfer, their memory demands can exceed the VRAM limits of consumer GPUs. A natural lower-cost alternative is to learn compact transport in the compressed space of a frozen VAE. However, nonlinear, lossy encoding complicates appearance endpoints, while spectral bias allows low-frequency structural residuals to dominate optimization and suppress high-frequency style gradients.

We propose WEAVE, a compact wavelet-routed framework for VAE-latent transport. A single-level Haar transform separates the latent into frequency bands. A lightweight velocity model learns source-aligned LL transport together with LH/HL motion, while stepwise statistical alignment injects target-style statistics into the high-frequency bands, including HH, without changing the LL base. This learned–analytic split makes the remaining latent transport learnable by a compact field.

We further introduce the IDT–TGT sandwich, which brackets transfer between an identity floor and a source-independent target-style anchor. WEAVE clears the identity baseline in both CLIP-S and DINO-S, occupying a competitive position on the style–content Pareto frontier. With only 1.04M trainable parameters, it trains in 1.4 minutes after latent caching on a single RTX 3060, showing that its efficiency does not arise from identity-like under-transfer.

Introduction

Recent diffusion and foundation-model editors provide strong priors for artistic style transfer, but their computational cost remains prohibitive for consumer deployment. Systems built on repeated diffusion denoising, inversion, or large image foundation models can require tens of gigabytes of GPU memory and minutes to hours of inference (Wu et al. 2025; Black Forest Labs 2025; Chung, Hyun, and Heo 2024; Hertz et al. 2024; Lu et al. 2024). A natural route to lower cost is to perform transfer in the compressed space of a frozen variational autoencoder, where spatial dimensions are smaller, representations can be cached, and the remaining transport can in principle be modeled by a compact network (Rombach et al. 2022). Yet VAE compression introduces a less visible obstacle: transformations that are simple in RGB space need not remain simple after nonlinear, lossy encoding. *VAE compression lowers spatial cost, but can make the transport itself substantially harder to learn.*

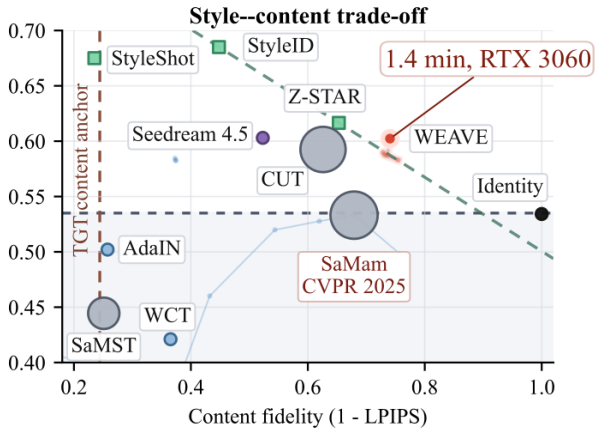


Figure 1: Style–content trade-off on D5, with mean style affinity defined as $(\text{CLIP-S} + \text{DINO-S})/2$. Bubble area denotes task-specific training time for trainable methods; training-free methods use a fixed marker size. WEAVE exceeds IDT on both style metrics and reaches higher mean style affinity than the similarly content-preserving SaMam operating point, at substantially lower training cost.

Direct latent endpoints are both less affine and harder for a matched predictor to learn than their pixel-space counterparts (Figure 4). Subband and gradient measurements attribute this difficulty to low-frequency residuals from unrelated reference geometry dominating high-frequency style motion.

We address this with **WEAVE**, a source-aligned, wavelet-routed latent flow that separates learned low- and mid-frequency transport from analytic high-frequency style injection: a compact velocity model predicts motion for the LL/LH/HL bands while stepwise statistical alignment supplies the remaining high-frequency style (Figure 3). This observation is broader than style transfer: latent representations optimized for reconstruction need not be compatible

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with low-capacity editing or conditional transport. WEAVE turns the observation into a solver design: it removes analytically tractable HF correction from the learned burden and reserves the compact field for the remaining base and oriented residual motion.

A compact model can also obtain favorable content-oriented scores simply by staying close to the input; we therefore evaluate WEAVE against an identity floor and a target-style copy anchor (**IDT–TGT sandwich**) to confirm its efficiency does not come from under-transferring. WEAVE clears the identity anchor under both DINO-S and CLIP-S while retaining substantially more content than comparable diffusion editors, at 1.04M trainable parameters and 1.4 minutes of training on a consumer GPU.

Contributions. Our contributions are threefold.

1. **Compact transport diagnosis.** We identify a compression–transport gap: VAE compression reduces spatial cost but can make simple appearance transformations substantially harder to learn. Endpoint residuals, matched convergence, and subband-gradient measurements provide diagnostics of a latent representation's transport compatibility.
2. **Wavelet-routed affine-fiber transport.** WEAVE uses the Haar-induced decomposition to assign source-aligned LL base motion, target-conditioned LH/HL vertical motion, and fiber-preserving high-frequency statistical alignment. This learned–analytic factorization makes residual latent transport learnable with a 1.04M-parameter field. The resulting collection-level transport prior can further register a new style by updating only its 0.098M memory entry.
3. **Valid transfer at minute-scale cost.** We introduce the IDT–TGT sandwich to verify that computational savings do not arise from identity-like behavior. WEAVE clears the identity anchor, occupies a competitive style–content Pareto position, and—restricted to sandwich-valid methods with fully reported timing—delivers about a $5\times$ lower end-to-end train+infer cost (counting the one-time latent-caching pass) than the fastest such baseline on the 750-image board.

Related Work

Classical NST replaces iterative optimization with feed-forward, attention-based, or style-bank transforms (Gatys, Ecker, and Bethge 2015, 2016; Huang and Belongie 2017; Li et al. 2017; Park and Lee 2019; Deng et al. 2022; Hong et al. 2023; Kwon et al. 2024; Park et al. 2020; Gim, Yoo, and Uh 2024; Zhang et al. 2024a; Zhang and Tang 2025; Chen et al. 2017). Unlike style-specific transformation banks, WEAVE uses memory only as a coarse prior and derives exemplar-specific conditioning from routed HF codes and moments.

Diffusion editors also act in frozen-VAE latents but still invoke a large denoiser at inference (Rombach et al. 2022; Hertz et al. 2022; Meng et al. 2021; Chung, Hyun, and Heo 2024; Hertz et al. 2024; Ye et al. 2023; Zhang et al. 2023; Kim, Kwon, and Ye 2023; Zhang et al. 2024b; Lu et al. 2024; Wu et al. 2025; Black Forest Labs 2025). WEAVE keeps the frozen VAE but removes the diffusion U-Net from transport, using a compact field plus an analytic update (validated as an SD1.5 plug-in).

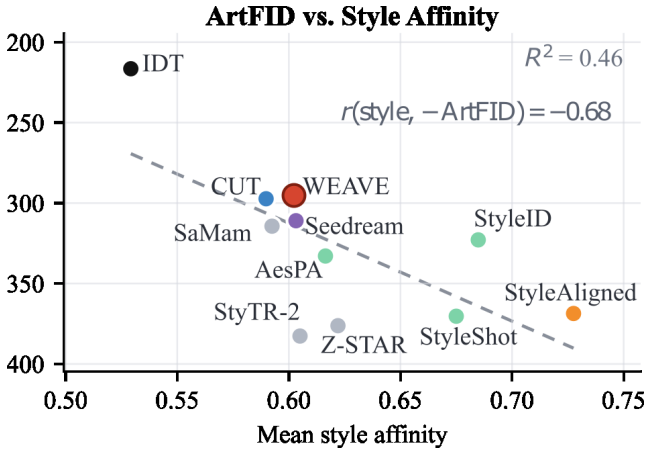


Figure 2: ArtFID vs. style affinity $(\text{CLIP-S} + \text{DINO-S})/2$ on D5 ($r(\text{style}, -\text{ArtFID}) = -0.68$). On D5, ArtFID favors strongly content-preserving operating points, including identity, and therefore should not be interpreted as a standalone ranking of style strength.

Flow-matching and Schrödinger-bridge methods learn transport between endpoint distributions (Liu, Gong, and Liu 2023; Lipman et al. 2022; Léonard 2014; De Bortoli et al. 2021; Shi et al. 2023; Liu et al. 2023; Su et al. 2023; Cuturi 2013; Peyré and Cuturi 2019). Recent latent variants reduce inference cost: LBM learns a latent bridge for fast image-to-image translation (Chadebec et al. 2025), and Latent Schrödinger Bridge reuses pretrained diffusion predictors to approximate unpaired transport with fewer evaluations (Kim, Kim, and Ye 2024). WEAVE keeps flow-matching supervision but reformulates exemplar-conditioned style endpoints and assigns each solver step a fiber-preserving analytic operator; its contribution is the learned–analytic responsibility split, not a new bridge objective.

Mamba-based state-space models give global receptive fields at linear cost for style transfer (Gu and Dao 2023): SaMam (Liu et al. 2025) uses a style-aware Mamba encoder–decoder over flattened spatial tokens without decomposing frequency subbands, while SaMST (Liu et al. 2024) decouples style learning from a compact convolutional transfer network. WEAVE instead decomposes the latent into an orthonormal Haar basis and routes LL/LH/HL to separate lightweight heads, reserving HH for analytic alignment—without a sequential backbone, a token-domain design, or a diffusion U-Net. LPIPS and ArtFID may reward content-preserving near-copies (Zhang et al. 2018; Wright and Ommer 2022), and no single metric captures both style affinity and content fidelity (Ioannou and Maddock 2024). On D5, style affinity correlates negatively with $-\text{ArtFID}$ ($r = -0.68$, Figure 2); we therefore report style and content separately and use the IDT–TGT sandwich as a validity frame rather than a style-strength ranking.

Spectral-bias and frequency-aware models separate scales

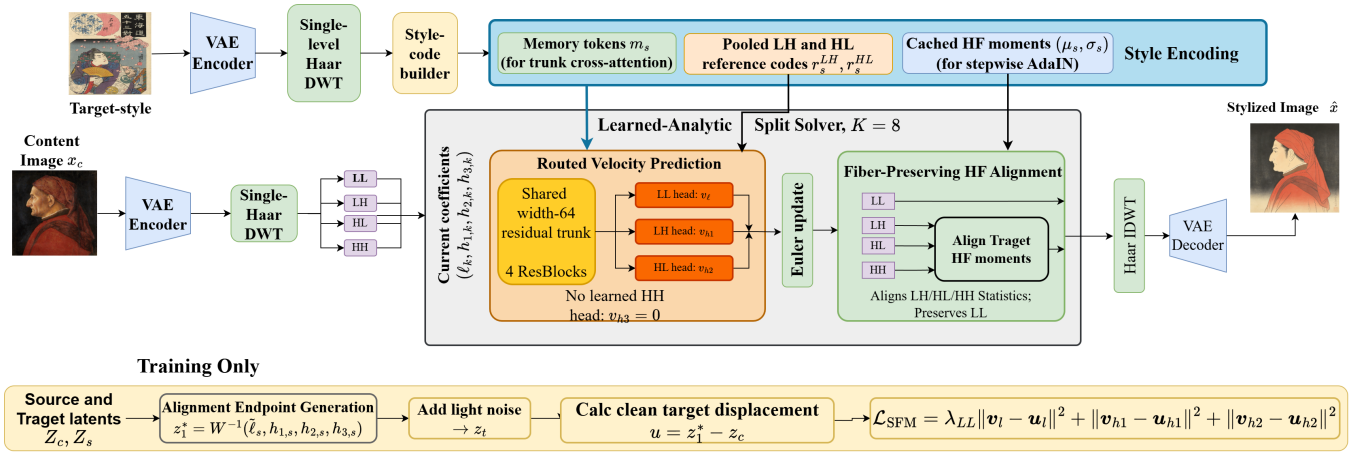


Figure 3: WEAVE architecture. At each of the $K = 8$ steps, a compact field updates LL/LH/HL; S_s then aligns HF statistics while preserving the updated LL base (no learned HH head). Style memory m_s is a learned lookup for registered style ID s; the target exemplar supplies pooled LH/HL codes to the learned routes and HF moments to S_s. Bottom: training-only endpoint and routed loss.

in objectives or generation heads (Rahaman et al. 2019; Xu, Zhang, and Luo 2022; Cai et al. 2021; Ren et al. 2026); wavelets give exact multi-band coordinates (Phung, Dao, and Tran 2023; Yoo et al. 2019; Daubechies 1992). Prior work uses wavelets for compression, high-resolution supervision, or one-shot style initialization (Sadat et al. 2024; Sigillo, He, and Comminiello 2025; Zhang et al. 2025; Wang et al. 2025); DGPST applies AdaIN–Wavelet initialization after semantic warping, whereas WEAVE routes learned LL/LH/HL motion with stepwise analytic HF alignment inside the solver, without a diffusion denoiser. Frequency-aware tokenizers such as the FA-VAE variants improve reconstruction by preserving or decoupling high-frequency information (Lin et al. 2023; Yin, Liu et al. 2025); WEAVE is complementary, freezing the tokenizer and asking whether the resulting latent geometry supports low-capacity conditional transport, which reconstruction fidelity alone does not determine.

Method

WEAVE formulates style transfer as wavelet-routed transport in VAE latent space (Figure 3): it source-aligns the LL endpoint, assigns LL/LH/HL motion to a compact velocity field, and handles the remaining high-frequency statistics analytically.

Why Direct Latent Transport Is Difficult

For an RGB map T, the normalized residual of the best pointwise affine fit is

$$R_q(T) = \min_{A,b} \frac{\|Aq(x_c) + b - q(T(x_c, x_s))\|_2^2}{\|q(T(x_c, x_s)) - q(x_c)\|_2^2}, \quad q \in \{I, \mathcal{E}\}. \tag{1}$$

Figure 4 shows $R_{\mathcal{E}} \gg R_I$ for color shift, AdaIN, WCT, and gamma maps, and a matched TinyUNet (same width; pixel at 256^2, latent at 32^2) converges far more slowly on direct latent

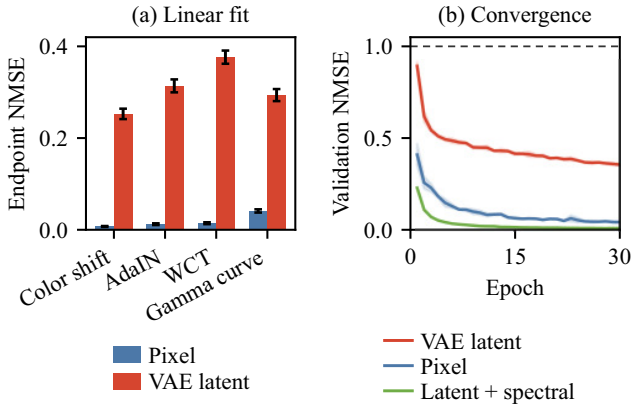


Figure 4: Pixel vs. VAE-latent endpoint fit. (a) Affine residual. (b) Matched TinyUNet: frequency-structured latents recover learnability.

endpoints than on pixel or frequency-structured latent endpoints. Shared MSE then lets low-frequency residuals from unrelated reference geometry dominate the update (Rahaman et al. 2019; Xu, Zhang, and Luo 2022; Cai et al. 2021). *We therefore reformulate the transport before increasing capacity.*

Haar-Induced Affine-Fiber Coordinates

A one-level orthonormal Haar map $\mathcal{W}(z) = (\ell, h_1, h_2, h_3)$ exposes LL and oriented HF bands (Daubechies 1992). With $P_L z = \mathcal{W}^{-1}(\ell, 0, 0, 0)$, $P_H = I - P_L$, and $\pi_L(z) = \ell$, the fiber over ℓ is the exact affine set

$$\mathcal{F}_\ell = \pi_L^{-1}(\ell) = \mathcal{W}^{-1}(\ell, 0, 0, 0) + \ker P_L, \tag{2}$$

Thus $\mathcal{Z} = \operatorname{im} P_L \oplus \ker P_L$. We use "affine fiber" only in the linear sense of Eq. 2: an operator coordinate split, not a semantic content/style partition. LL may still contain style, and HF may contain geometric edges.

Source-Aligned LL Endpoint

Let $z_c{=}\mathcal{E}(x_c)$ and $z_s{=}\mathcal{E}(x_s)$ from the frozen SD1.5 VAE (Rombach et al. 2022). Direct $z_c{\to}z_s$ supervises unrelated base geometry in ℓ_s. WEAVE replaces the base destination by

$$\begin{aligned} \tilde{\ell}_s &= (1-\alpha)\ell_c + \alpha\,\mathrm{AdaIN}(\ell_c, \ell_s), \qquad \alpha = 0.3, \\ z_1^\star &= \mathcal{W}^{-1}(\tilde{\ell}_s, h_{1,s}, h_{2,s}, h_{3,s}), \end{aligned} \tag{3}$$

keeping target HF coordinates. Training uses a lightly noised rectified path

$$z_t = (1-t)z_c + tz_1^\star - \sigma\sqrt{t(1-t)}\,\epsilon, \qquad u = z_1^\star - z_c, \tag{4}$$

with $t{\sim}\mathcal{U}([0,1])$, $\epsilon{\sim}\mathcal{N}(0,I)$, $\sigma{=}0.02$; supervision is the clean displacement u (noise off at inference).

Learned Base and Vertical Transport

The field predicts only base and LH/HL vertical velocities,

$$v_\theta(z_t, t, s) = (v_\ell,\, v_{h_1},\, v_{h_2},\, 0), \tag{5}$$

Because $\mathcal{W}$ is orthonormal, Parseval gives an exact subband decomposition of any latent residual:

$$\begin{aligned} \|z{-}z'\|_2^2 = \|\ell{-}\ell'\|_2^2 + \|h_1{-}h_1'\|_2^2 \\ + \|h_2{-}h_2'\|_2^2 + \|h_3{-}h_3'\|_2^2. \end{aligned} \tag{6}$$

Under a shared MSE on z, a large LL residual can dominate the total gradient and starve the HF heads. Subband routing makes this frequency-scale imbalance explicit and controllable: spectral matching with $\lambda_{LL}{=}0.3$ (results remain stable at 1.0; Table 2)

$$\mathcal{L}_{\mathrm{SFM}} = \mathbb{E}_t\big[\lambda_{LL}\|v_\ell - u_\ell\|_2^2 + \|v_{h_1} - u_{h_1}\|_2^2 + \|v_{h_2} - u_{h_2}\|_2^2\big] \tag{7}$$

weights LL explicitly, while HH never enters the learned gradient (analytic path only). A width-64 residual trunk with separate $1{\times}1$ LL/LH/HL heads implements v_θ; no HH head is learned. Style enters through one learned 256×384 memory entry per registered style. Reference exemplars independently provide pooled LH/HL codes to the learned routes and HF moments to S_s. Unseen styles clamp the memory path but retain both reference-derived routes. Registration appends a new memory entry and optimizes only its 0.098M parameters while freezing the remaining model (Table 5).

Fiber-Preserving Statistical Alignment

For $q{=}P_H(z)$ and $q_s{=}P_H(z_s)$, channel-wise AdaIN (Huang and Belongie 2017)

$$\mathcal{A}(q; q_s) = \frac{q - \mu(q)}{\sigma(q) + \varepsilon}\,\sigma(q_s) + \mu(q_s) \tag{8}$$

defines the fiber-preserving map

$$\begin{aligned} S_s(z) &= P_L z + \Psi_s(P_H z), \\ \Psi_s(q) &= (1-\beta)q + \beta\,\mathcal{A}\big(q; (1+\eta)q_s\big). \end{aligned} \tag{9}$$

Because $q = P_H z = \mathcal{W}^{-1}(0, h_1, h_2, h_3)$ is the reconstructed HF residual, each local Haar block has zero mean. Hence $\mu(q) = \mu(q_s) = 0$ and

$$\mathcal{A}(q; q_s) = q \cdot \frac{\sigma(q_s)}{\sigma(q) + \varepsilon},$$

which remains a channel-wise scaling in $\ker P_L$.

Proposition 1 (Base preservation)**.** *$\pi_L(S_s(z)) = \pi_L(z)$ for all z, i.e. $S_s : \mathcal{F}_\ell \to \mathcal{F}_\ell$.*

Proof. $P_H z \in \ker P_L$ and the scaling property of $\mathcal{A}$ imply $\Psi_s(P_H z) \in \ker P_L$, so $\pi_L(S_s(z)) = \pi_L(P_L z) = \pi_L(z)$. □ □

Thus S_s injects HF statistics (and supplies HH) without moving the current LL base; applying it after every Euler step makes alignment part of the dynamics, not post-processing.

Learned–Analytic Split Solver

Inference alternates a learned Euler step and S_s:

$$z^{k+1} = S_s \circ \Phi_\theta^{1/K}(z^k), \qquad k = 0, \ldots, K-1, \tag{10}$$

with $K{=}8$. Training minimizes Eq. 7 on cached latents; the model has 1.04M trainable parameters plus the frozen 84M VAE. Details are in the supplement.

Experiments

Implementation Details

Datasets. All benchmarks are built from publicly available WikiArt images. D5 uses the five WikiArt families whose identity/target-style CLIP-S gap is largest (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e). Training uses 5,000 images (1,000 per family) and testing uses 150 images (30 per family); each test image is paired with one reference from every family, yielding 750 content–style pairs at $512{\times}512$ resolution. P2A is the standard photo-to-art benchmark evaluated at 256×256. To control for the CLIP-gap-based selection of D5, R5-WikiArt uses five styles drawn at random from WikiArt (Cubism, Expressionism, Pop Art, Romanticism, Symbolism). As on D5, each of the 150 content images is paired with one reference from every family, yielding 750 pairs at 512×512. It is distinct from Other5 below (a different five-family held-out set used for zero-shot conditioning and memory-registration experiments). For each request, the source content image and a target-style reference image are given; the task is to render the content in the target style.

Metrics. DINO-S measures target-style affinity as the maximum cosine similarity between DINOv2 (Oquab et al. 2023) CLS embeddings of the generated image and the CLS embeddings of the target-style reference pool (up to 30 reference images per style), following the standard photo-to-art evaluation protocol. CLIP-S reports the cosine similarity between CLIP (Radford et al. 2021) image features of the generated image and a target-style prototype formed as the L2-normalized mean CLIP feature over a fixed 16-image reference pool per style; the prototype is precomputed once

Method	D5-512				P2A-256				R5-WikiArt				Params	Train/Infer
	D-S	C-S	LP↓	D-C	D-S	C-S	LP↓	D-C	D-S	C-S	LP↓	D-C	M	min
IDT	0.419	0.640	0.000	1.000	0.356	0.618	0.000	1.000	0.433	0.731	0.000	1.000	0	–/–
TGT	1.000	0.855	0.756	0.220	1.000	0.835	0.800	0.169	1.000	0.796	0.747	0.230	0	–/–
SD-Turbo	0.484	0.693	**0.003**	**0.922**	0.341†	0.674	0.603	0.279	0.505	0.767	0.449	0.538	0^{+1100}	free/5
StyleAligned	**0.675**	0.780	0.869‡	0.239	**0.612**	0.768	0.786	0.310	**0.649**	**0.824**	0.829‡	0.315	0^{+1100}	free/77
Z-STAR	0.449	0.784	0.347	0.549	0.514	**0.786**	0.332	0.526	0.514	**0.822**	0.384	0.526	0^{+1100}	free/180
StyleShot	**0.563**	**0.787**	0.765‡	0.377	**0.552**	**0.774**	0.713	0.335	0.484	0.787	0.795‡	0.380	0^{+2400}	free/306
StyleID	0.548	**0.822**	0.552	0.416	0.256†	0.648	0.691	0.147‡	**0.571**	0.815	0.493	0.374	0^{+1100}	free/63
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710†	0.620	**0.795**	7.0	323/5
SaMST	0.271†	0.618†	0.749	0.145‡	0.501	0.709	0.398	0.838	0.461	0.667†	0.612	0.615	8.3	39.5/10
SaMam	0.477	0.702	**0.248**	**0.812**	0.505	0.677	**0.205**	**0.870**	0.503	0.712†	**0.227**	**0.925**	8.8	436/17.6
StyTR-2	0.501	0.709	0.596	0.747	0.515	0.722	0.495	0.555	0.533	0.759	0.553	0.737	35^{+13}	1440/38
AesPA-Net	0.520	0.724	0.548	0.601	0.498	0.744	0.571	0.522	0.545	0.782	0.409	0.728	11^{+20}	6.1/23
Seedream	0.486	0.720	0.477	0.739	0.517	0.752	**0.227**	0.785	0.504	0.742	0.486	0.725	–	API/–
Latent-WCT	0.362†	0.673†	0.441	0.559	0.338†	0.738	0.585	0.386	0.414†	0.710†	0.444	0.553	0^{+84}	free/3.67
WEAVE	0.492	0.713	0.259	0.810	0.480	0.668	0.312	**0.861**	0.523	0.775	**0.290**	0.772	**1.04**$^{+84}$	**1.4/1.22**

Table 1: Main comparison (↑/↑/↓/↑ for D-S/C-S/LP/D-C). Best two non-anchor results are bold; double underline marks the best. †: the style metric does not exceed its IDT value; ‡: the content metric falls beyond the TGT reference. Params are in millions, with frozen parameters shown as superscripts; Train/Infer gives training and inference time in minutes for 750 outputs on an RTX 3060 unless noted. Trainable baselines use the same D5 split and evaluation board.

per target style and shared across all 750 requests, so CLIP-S measures absolute style affinity rather than similarity to a particular reference image. LPIPS measures perceptual content distance between the generated image and the source content image (Zhang et al. 2018). DINO-C reports DINOv2 cosine similarity between the generated image and the source, giving a complementary content-preservation view. IDT (identity transfer) and TGT (target-style copy) are dataset-specific anchors: IDT returns the unchanged source image (no-op floor), while TGT returns a target-style reference as the opposite extreme (high style affinity, low content). They are empirical anchors, not mathematical bounds. IDT/TGT are evaluated using the same metric-specific reference protocol as the methods they bracket (D5: 750 pairs; P2A IDT: cross-style transfer pairs with LPIPS=0 and DINO-C=1 by construction). We mark methods that do not exceed IDT on style (†) or fall outside TGT on content (‡) in Table 1.

Configuration. Training uses α=0.3, λ_{LL}=0.3, and σ=0.02 with evaluation-free internal stopping. Inference uses β=2.0, η=0.1, and K=8 across all boards; complete sensitivity results are provided in the supplement.

Baseline training protocol. Trainable baselines use the same D5 split and 750-pair evaluation board; timings are measured on RTX 3060-class hardware where available. Training-free editors use frozen public checkpoints or API outputs. D5, P2A, and R5 use collection-specific WEAVE models with shared solver defaults.

Comparison with State-of-the-Art Methods

D5 quality–compute operating point. WEAVE clears IDT on both style metrics and reaches higher mean style affinity than SaMam at similar content fidelity, with 1.04M trainable parameters and substantially lower train/infer cost. Editors with higher raw style affinity incur considerably larger content loss, as contextualized by TGT. D5 uses the oriented target-HF route; P2A and R5 retain the same Haar/endpoint/stepwise-alignment factorization with the base architecture.

Reading the sandwich. IDT identifies identity-like under-transfer, whereas TGT marks target-copy-like content loss. WEAVE clears both IDT style anchors without approaching TGT content loss; StyleAligned attains higher DINO-S but crosses the TGT reference in LPIPS.

P2A and R5. WEAVE remains inside the IDT–TGT bracket on P2A. On R5, it improves DINO-S over IDT, CUT, and SaMam while preserving more content than StyTR-2 and AesPA-Net, supporting the same operating point beyond the CLIP-gap-selected D5 styles.

Cost. Latent caching, training, and 750-image inference take approximately 3.0, 1.4, and 1.2 minutes, respectively, giving a 5.6-minute first-run budget on one RTX 3060. Under the same five-style protocol, this is about $10\times$ lower than StyleID's inference-only cost and $80\times$ lower than our SaMam retrain's train+infer cost. The complete pipeline fits within 12 GB VRAM.

Ablation Study

Table 2 reports a transport-responsibility ablation: each variant removes or reassigns one responsibility in Align–Route–Assign. Source alignment matters most for content: replacing it with a direct target endpoint increases LPIPS by 0.05 while leaving DINO-S nearly unchanged. HF conditioning and stepwise AdaIN each independently contribute to style

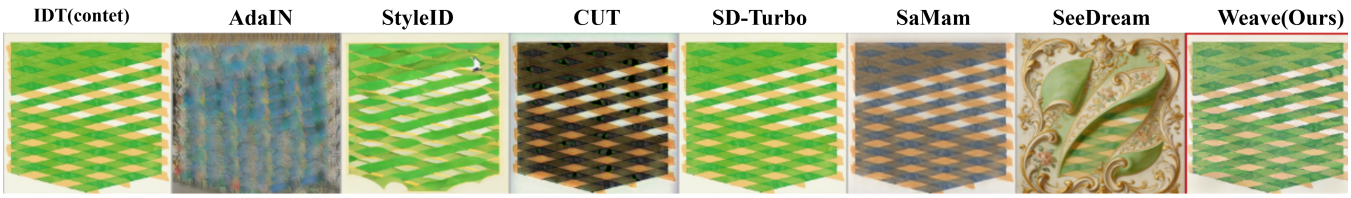


Figure 5: High-content cross-domain transfer from Minimalism to Rococo. Several baselines remain close to identity, whereas some diffusion editors alter the source structure. WEAVE preserves the source topology while shifting local color and texture toward the target, illustrating the operating point quantified by the IDT–TGT analysis.

affinity; final-only AdaIN collapses content. Pure S_s supplies most style movement; the learned field adds further style affinity and carries the LL and oriented residual motion. A learned HH head and λ_{LL}=1.0 give only marginal changes, motivating the simpler default.

Per-step diagnostics confirm that S_s supplies the dominant HF motion while the learned field carries LL and smaller oriented residual motion; full trajectories are provided in the supplement.

Variant	D-S↑	C-S↑	LP↓	D-C↑
WEAVE	0.492	0.713	0.259	0.810
w/o AdaIN	0.482	0.701	0.226	0.846
final-only AdaIN	0.484	0.729	0.395	0.666
w/o HF cond.	0.483	0.716	0.284	0.813
λ_{LL}=1.0	0.491	0.718	0.263	0.787
direct endpoint	0.489	0.717	0.314	0.797
learned HH	0.493	0.716	0.267	0.806
zero velocity	0.475	0.703	0.258	0.856

Table 2: Ablation on 750 D5 pairs. Zero-velocity disables the learned field (pure S_s).

Generalization and Robustness

We test the formulation across random seeds, VAE representations, held-out style families, and a frozen editor.

Multiple seeds. The evaluation-free stopping rule selects epochs 4, 4, and 3 for seeds 42, 7, and 123. On a fixed latent probe, it stops at the first target-HF gate expansion accompanied by at least a 35% contraction of the shared-trunk LL/HF gradient-norm ratio; no decoded image or image-quality metric is used. Algorithm S1 gives the complete rule.

Representation robustness. Table 3 replaces the frozen SD1.5 EMA VAE with FLUX.1, FLUX.2, and Qwen-Image, retaining the same one-level Haar-routed factorization and evaluation-free internal-dynamics selection, adapting channel projections to each latent width, and using the learned-HH portability variant described in the supplement. All three clear IDT on style with competitive content; FLUX.2 matches SD1.5 DINO-S while improving CLIP-S, LPIPS, and DINO-C. *The factorization is not tied to one tokenizer*, though the selected epoch shifts with latent geometry. We retain Haar for orthogonality and cost.

Frozen VAE	D-S↑	C-S↑	LP↓	D-C↑
SD1.5 EMA (main)	0.492	0.713	0.259	0.810
FLUX.1	0.489	0.713	0.256	0.780
FLUX.2	0.492	0.717	0.239	0.873
Qwen-Image	0.478	0.714	0.263	0.825

Table 3: Cross-VAE D5 under internal-dynamics selection (same evaluation-free rule as SD1.5; SD1.5 is seed-42 main).

Conditioning generalization. Across resampled reference pools, WEAVE retains a positive board-level DINO-S margin over IDT. On the five held-out Other5 families, the D5 checkpoint gives the best CLIP-S and content metrics, while SaMam is 0.015 higher in DINO-S (Table 4), showing that the reference-conditioned path generalizes beyond the registered style IDs.

Method	D-S↑	C-S↑	LP↓	D-C↑
WEAVE	0.5313	**0.7742**	**0.2802**	**0.7587**
SaMam	**0.5466**	0.7577	0.3990	0.7369
SaMST	0.3913	0.7485	0.6213	0.2680

Table 4: Zero-shot transfer on Other5 using the D5 checkpoint with the style-ID path clamped. All five target families are held out from D5 (750 pairs).

Memory-only style registration. Starting from the D5 checkpoint, we append five Other5 entries and optimize only their 0.49M parameters for 3,000 steps. Registration takes 98 s, improves all four metrics over zero-shot, and matches full-field fine-tuning within 0.003 DINO-S and 0.001 CLIP-S (Table 5). One-image registration already approaches the 30-image result (Table S6).

Operator portability. Applying the same analytic alignment to a frozen SD1.5 editor improves CLIP-S and DINO-S on 600 cross-style pairs without updating model weights, with only 4.9 ms overhead (Table 6; settings in the supplement).

Hyperparameter robustness. WEAVE is stable for $K \in \{2, 4, 8, 16\}$ and $\beta \in [1, 2]$; wavelet level, local alignment, and $(\alpha, \lambda_{LL}, \eta)$ sensitivities are reported in Tables S7–S11.

Setting	Adapt. M/time	D-S all/off	C-S	LP↓	D-C
Zero-shot	0/–	0.531/0.475	0.774	0.280	0.759
Mem.-only	0.49/98s	0.562/0.499	0.784	0.265	0.812
Full FT	1.53/4m	0.565/0.500	0.785	0.247	0.821

Table 5: Registration of five Other5 styles from the D5 checkpoint. Adapt. M/time denotes the parameters updated during registration, in millions, and registration time. All/off reports DINO-S over all 750 pairs and the 600 off-diagonal pairs. Each style uses 30 registration images and a fixed 3,000-step budget.

SD1.5 editor	D-S↑	C-S↑	LP↓	D-C↑
Base	0.3974	0.7221	0.3591	0.6589
+ wavelet alignment	**0.4080**	**0.7366**	0.3630	0.6531

Table 6: Frozen SD1.5 plug-in on 600 cross-style D5 pairs (no SD weights updated).

Discussion

WEAVE exposes a compression–transport gap: a tokenizer may reconstruct accurately while mapping a simple RGB transformation to a nonlocal, non-affine latent displacement. Reconstruction fidelity therefore does not determine the learnability of a downstream vector field. Endpoint non-affinity, subband displacement energy, and route-specific gradient balance provide complementary diagnostics of a latent representation's transport compatibility.

WEAVE also illustrates an invariant-preserving alternative to increasing neural capacity. For a projection P and an update Ψ satisfying $\Psi(\ker P) \subseteq \ker P$, the operator $S(z) = Pz + \Psi((I - P)z)$ preserves Pz. Rather than only reweighting the loss of a shared predictor, this construction removes a tractable component from the learned update and composes it with the numerical solver. Haar coordinates provide one inexpensive realization; other tasks may admit different projections and analytic corrections.

Implications for latent diffusion and flow models. These results suggest separating representation, transport, and solver design rather than treating the latent space as a neutral computational compression. First, tokenizer selection may consider the geometry of task-relevant endpoint displacements in addition to reconstruction quality. Second, when a known correction acts in a complementary subspace—for example, a color, frequency, or moment constraint—it can be interleaved with learned denoising or flow steps instead of being regressed by the same high-capacity predictor. Such a split reduces what the learned predictor must represent while preserving designated coordinates by construction. This perspective is complementary to fast samplers: samplers reduce how often a learned field is evaluated, whereas operator splitting reduces the responsibility assigned to that field.

The cross-VAE experiments show that the routed factorization survives several latent geometries, while the frozen-SD1.5 plug-in isolates the portability of the analytic operator. Memory-only registration further suggests a separation between reusable collection-level dynamics and lightweight condition-specific adaptation. These results motivate transport-aware tokenizer selection and learned–analytic solvers, but do not by themselves establish improvements to full-scale diffusion training.

Limitations

WEAVE attains a stronger operating point on D5-512 than on P2A-256, but the datasets differ, preventing a controlled conclusion about the effect of resolution or retained high-frequency structure. Matched evaluations across 256, 512, and larger resolutions are needed to determine how the routed transport scales with latent spatial size. Style coverage is evaluated on five-style collections and five registered held-out families; scaling the shared transport prior and memory bank to substantially larger and more diverse style sets remains to be established. Finally, cross-VAE results demonstrate portability across latent representations, and the frozen-SD1.5 plug-in isolates the analytic operator, but neither establishes gains across broader diffusion backbones or end-to-end diffusion/flow training. Testing learned–analytic splitting at those scales and model regimes is an important direction for future work.

Conclusion

WEAVE shows that minute-scale style transfer in a frozen VAE latent space can be achieved by reformulating the transport before allocating additional model capacity. Source-aligned endpoints remove unrelated target geometry from the LL destination, wavelet routing exposes the frequency-scale imbalance of direct latent supervision, and fiber-preserving HF alignment assigns a tractable part of the transfer to an analytic solver step. Together, these choices reduce the remaining transport burden to a 1.04M-parameter field.

The resulting model exceeds the identity anchor on both style metrics, trains in 1.4 minutes after latent caching, and supports 98 s memory-only registration of new style families. Cross-VAE experiments, held-out-style evaluation, and the frozen-SD1.5 plug-in further suggest that the transferable principle is not a particular compact architecture, but the underlying division of responsibilities among the latent representation, learned field, and analytic update. The IDT–TGT sandwich complements the cost accounting by verifying that the observed efficiency does not arise from identity-like under-transfer or target-copy-like content loss.

More broadly, WEAVE distinguishes three properties that are often conflated in latent modeling: compression efficiency, reconstruction fidelity, and transport compatibility. A latent representation may be compact and reconstruct accurately while still making a downstream conditional displacement difficult for a small predictor to learn. The central lesson is therefore to diagnose and reformulate the transport, and to assign tractable components analytically, before increasing learned capacity.

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